ENT

2026-27

CONCEPTS | CLINICALS | MCQS

-Small Subject. High Yield Impact.
-Audiograms, instruments & key clinical signs.
-Rapid revision of frequently asked topics.

DR. KEYUR GANDHI DR JITENDRA

Aspira Edge

ENT

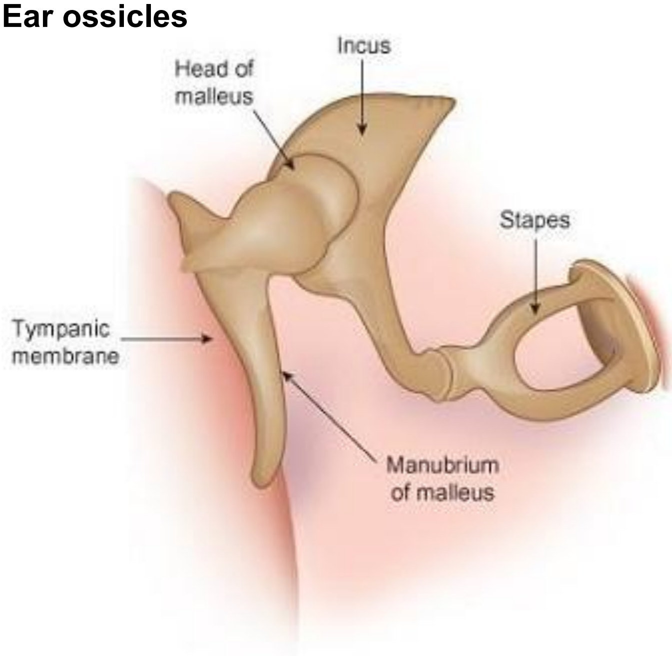
Embryological development of ear **FMGE 2024**
Development of pinna → 1st & 2nd branchial arch
Tragus → 1st branchial arch

Development of EAC → from 1st brachial arch
Development of eustachian tube and middle ear → first branchial pouch
Development of TM → all 3 layers (ectoderm + endoderm + mesoderm)

Development of ossicles -
malleus and incus → 1st brachial arch
stapes → 2nd brachial arch (foot plate - neuroectoderm)

Development of cochlea → Neuroectoderm

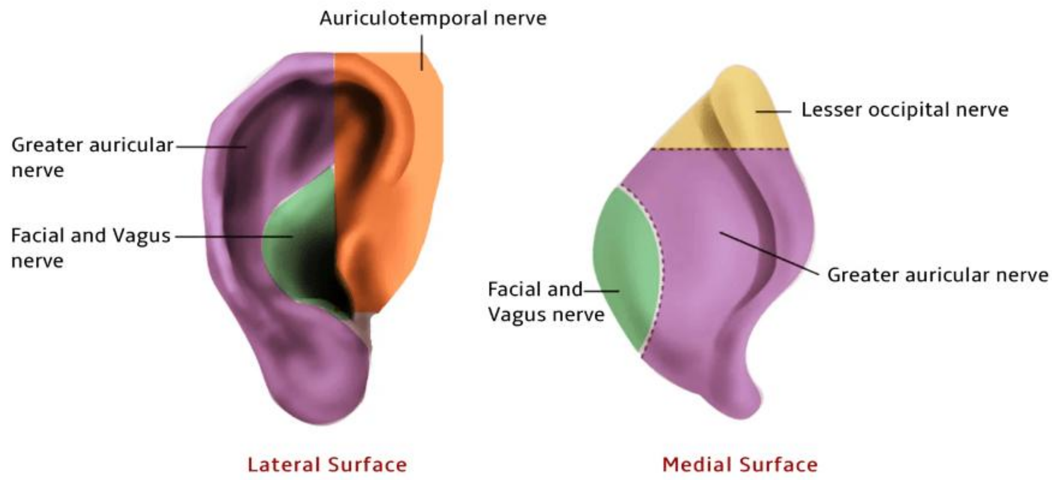
Ear ossicles **NEETPG 22**



size → malleus>incus>stapes

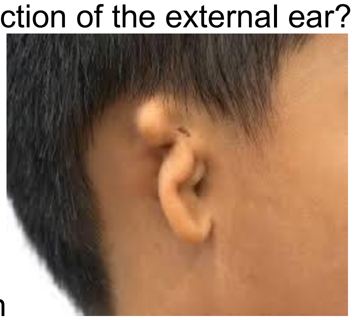
NERVE SUPPLY OF EAC **NEETPG 24**

1. Auriculotemporal nerve - It supplies the anterior wall and roof of EAC.
2. Auricular branch of vagus (arnold's or alderman's nerve) - Posterior wall and floor. Stimulation of this nerve causes coughing.
3. Sensory division of facial nerve - It supplies the posterosuperior part of EAC
4. Greater auricular nerve (C2-C3) → main supply
5. Lesser occipital nerve → upper part of medial surface



Frequently asked question

Q) A child is born with unilateral microtia. What is the ideal age for surgical reconstruction of the external ear?



A. At birth
B. 2 years
<u>C. 6 years and above</u>
D. Anytime

Explanation: Microtia reconstruction = wait till rib cartilage matures →ideally at 10–12 year of age

DISEASES OF EAC

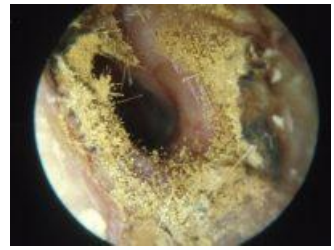
1.Malignant otitis externa

FMGE 2024,22

infection of bone of EAC
Risk factor - diabetes,old age, ↓ immunity
organism- pseudomonas
on examination → granulation in EAC
c/f- severe ear ache
facial palsy
blood stained purulent ear discharge

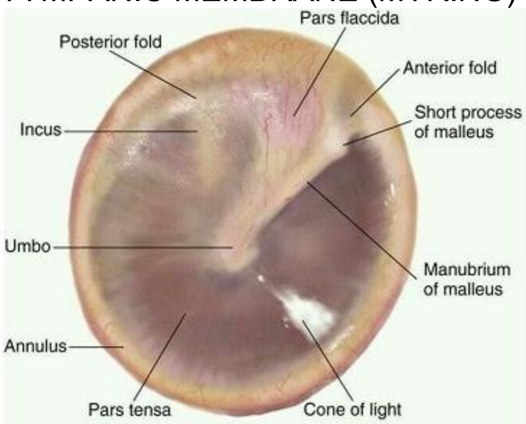
Inv- HRCT temporal bone
Trt- 3rd generation cephalosporins

2.Otomycosis



organism → aspergillus niger
o/e → wet newspaper appearance

TYMPANIC MEMBRANE (MYRING)

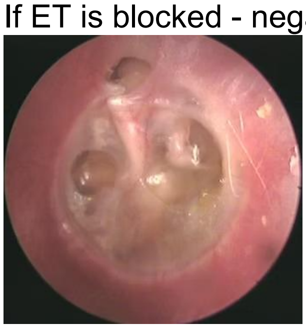


TM has 2 parts

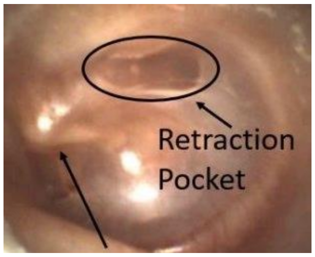
Pars flaccida	Pars tensa
stronger part of TM	weaker part of TM

Eustachian tube
main function → middle ear ventilation
which muscle open eustachian tube during swallowing → Tensor velli palatini
ET function test → Tympanometry → Best test

CHOLESTEATOMA
If ET is blocked - negative pressure (vacuum) in middle ear leads to RETRACTED TM.



Dull in appearance - too much retraction will form RETRACTION POCKET



The most common site of retraction pockets is; Pars flaccida because it lacks a middle layer.

If retraction pocket allows progress it will rupture & through that perforation skin will into middle ear, skin in middle ear is called CHOLESTEATOMA.

GLUE EAR (Serous otitis media) **NEETPG 22,FMGE 2025,22**

1. Collection of thick STERILE glue like fluid in middle ear.
2. Cause- Adenoid hypertrophy (MCC) /Nasopharyngeal ca.

c/f -
Conductive hearing loss
Heaviness in ear (no pain)
Poor school performance

Rx: Surgery: myringotomy (Antero-inferior quadrant) + grommet insertion (middle ear ventilation tube) +/- Adenoidectomy

Myringitis bullosa **FMGE 2023**

cause - HSV,Mycoplasma,E multiforme
Painful
self limiting

Myringosclerosis
Hyaline degeneration of middle fibrous layer of TM

MIDDLE EAR (TYMPANUM)
3 parts:
1. Mesotympanum
● It is covered by TM

2. Epitympanum
● It's the upper most part of the middle ear. ● It is covered by bone called: SCUTUM

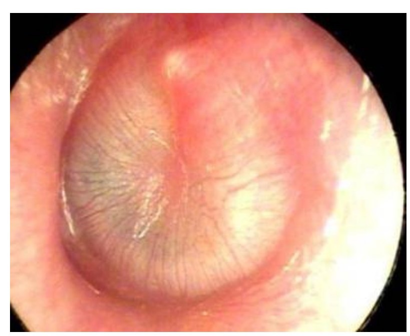
3. Hypotympanum
● Lowest part of middle ear
● It is covered by bone

NEETPG 23

Sensory supply of middle ear → tympanic branch of glossopharyngeal nerve (jacobson nerve)

Referred otalgia
→ pain via glossopharyngeal nerve
Seen in → Tonsillectomy
Tonsillitis
Eagle syndrome

ACUTE SUPPURATIVE OTITIS MEDIA (ASOM)
Definition: Infection of the middle ear mucosa.
Common organism: Streptococcus pneumoniae
Clinical Features:
1.Severe earache
2.Red, bulging tympanic membrane (TM) with dilated vessels
3.Cart-wheel appearance on otoscopy



Treatment:

Red, bulging TM → Do Myringotomy (Postero-inferior quadrant)
If untreated → TM perforation → persists >3 months → SAFE CSOM (Chronic Suppurative Otitis Media)

Remember:
ASOM → Untreated → Permanent perforation → Safe type CSOM

SAFE CSOM **NEET PG 25,FMGE 2025**
Chronic infection of the middle ear with permanent central perforation in the pars tensa (lower part) of the tympanic membrane.
Clinical Features:
1.Ear discharge – thin, non-foul smelling, not blood-stained
2.Conductive hearing loss
3.Ossicular erosion may occur — most commonly the incus

AUSTIN CLASSIFICATION OF OSSICULAR EROSION

Type A	Incus - ,Malleus + ,stapes +
Type B	Incus - ,Malleus + ,stapes -
Type C	Incus - ,Malleus - ,stapes +
Type D	Incus - ,Malleus - ,stapes -

- = absent += present

Remember:
Safe CSOM = Central perforation, safe for life but not for hearing!

Trt-
Surgery is the main treatment.
1.Myringoplasty: Repair of tympanic membrane (TM) perforation using a temporalis fascia graft (most commonly used).
Most common approach is post aural → wilde post aural incision
We have to check ossicles before surgery

2.Type III Tympanoplasty (Columella Tympanoplasty): Done when incus and malleus are eroded but the stapes are intact. The TM graft is placed directly over the stapes, forming a sound-conducting column.
This procedure is also called Myringostapediopexy.

3.ossicular replacement prosthesis

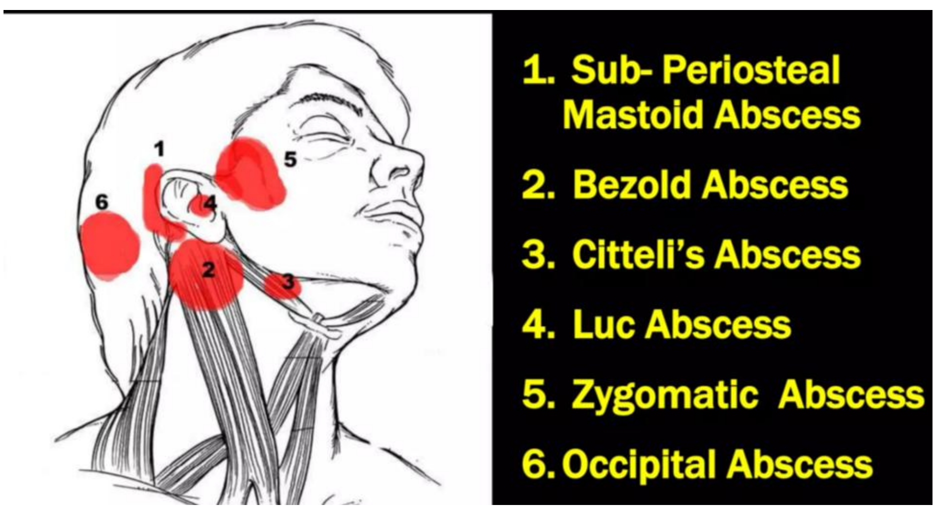
ACUTE MASTOIDITIS

Definition: Infection of mastoid air cells, usually a complication of ASOM or CSOM.
Clinical Features:
Pain and swelling behind pinna
Red, shiny mastoid skin → “Ironing of mastoid surface” (first sign)
Profuse ear discharge that refills after cleaning → “Reservoir sign”
Pulsatile otorrhea +
Investigation: X-ray mastoid: Schuller’s & Towne’s views

Treatment:
IV antibiotics for 48 hrs → no response → Cortical Mastoidectomy (Schwartz operation)

Gradinego syndrome / Petrositis **FMGE 2024**
Triad → G
E- ear discharge
R-retro orbital pain due to 5th CN
D- diplopia → 6th CN
Inv- HRCT temporal bone

Abscess formation in mastoiditis **NEETPG 22,FMGE 2025**



UNSAFE CSOM
Cholesteatoma → ectopic presence of skin in middle ear cleft
→ most common site is Prussak space of epitympanum

Types of cholesteatoma

Primary acquired cholesteatoma	Most common origin of cholesteatoma Due to retraction pocket
Secondary acquired cholesteatoma	Due to marginal perforation
Tertiary acquired cholesteatoma	Accidental implant of skin during surgery
Congenital cholesteatoma	Pearly white mass behind tympanic membrane

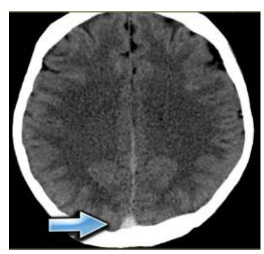
c/f-
Ear discharge - foul smell and blood stained
CHL
Sx-
MRM- modified radical mastoidectomy

Labyrinthine fistula
Fistula on lateral SCC of inner ear
c/f - vertigo
Fistula sign seen with seigelisation → nystagmus,vertigo

False positive fistula sign	False negative fistula sign
Congenital syphilis Meniere disease	Fistula in dead labyrinthine Fistula covered by cholesteatoma

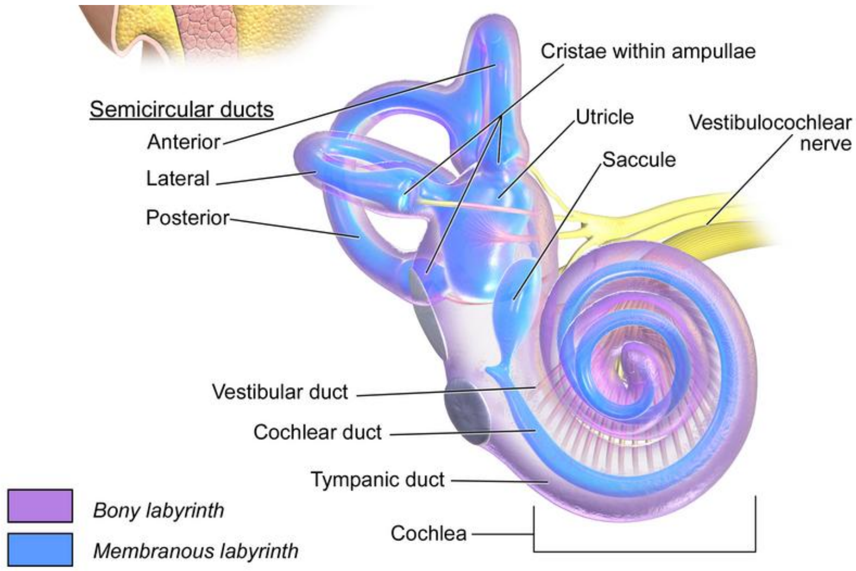
Trt -
MRM + cover the fistula with temporalis fascia

SIGMOID SINUS THROMBOSIS



Intracranial complication
c/f-
Headache
Spiky fever → picket fence fever
Grissenger sign
CT brain → Delta sign
Trt - MRM + open the sinus plate → clear thrombus

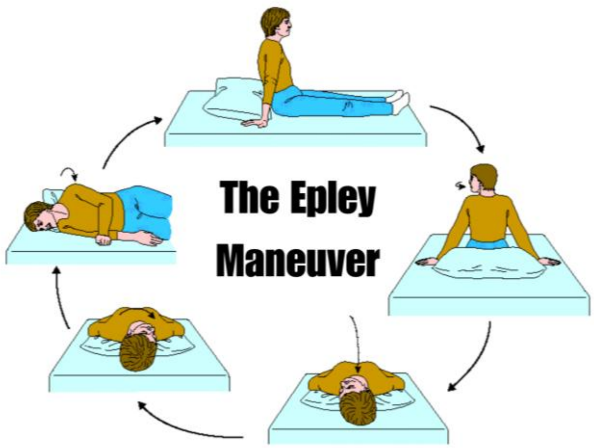
INNER EAR **FMGE 2024,23**
COCHLEA- Organ of corti for hearing.
UTRICLE & SACCULE- Macula for liner balance. (U- horizontal,S- vertical balance)
SEMICIRCULAR CANALS - Crista for angular balance.
Function – Balance & Equilibrium

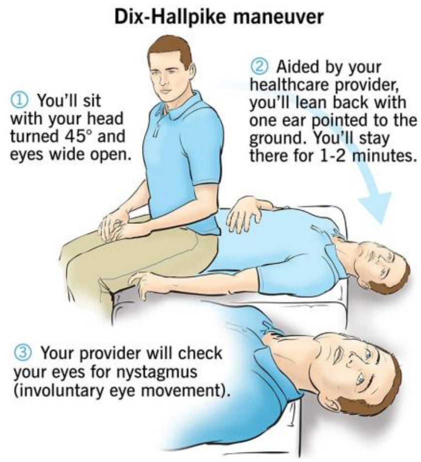


TOTAL 3 PARTS OF COCHLEA
1. Scala vestibuli 2. Scala media 3. Scala tympani

Membrane between Scala vestibuli & Scala media is called: REISSNER'S MEMBRANE
Membrane between scala media & scala tympani called: BASILAR MEMBRANE.
Above scala tympani there is a structure called: ORGAN OF CORTI.
Organ of corti is covered by TECTORIAL MEMBRANE

BENIGN PAROXYSMAL POSITIONAL VERTIGO
Causes: Displaced otoconia (utricle saccule → SCC)
Common in female and most common causes of peripheral vertigo
Most commonly involved semicircular canal (posterior semicircular canal)
C/F: Vertigo for a few seconds on changing head position
Diagnostic → Dic hallpike manuvere
Trt- Epeleys manuvere





FITZGERALD BITHERMAL CALORIC TEST **FMGE 2023**

For lateral SCC only
Warm water - 44°c
Cold water - 30°c

Direction of nystagmus - CO WS
Cold - opposite
Warm - same side

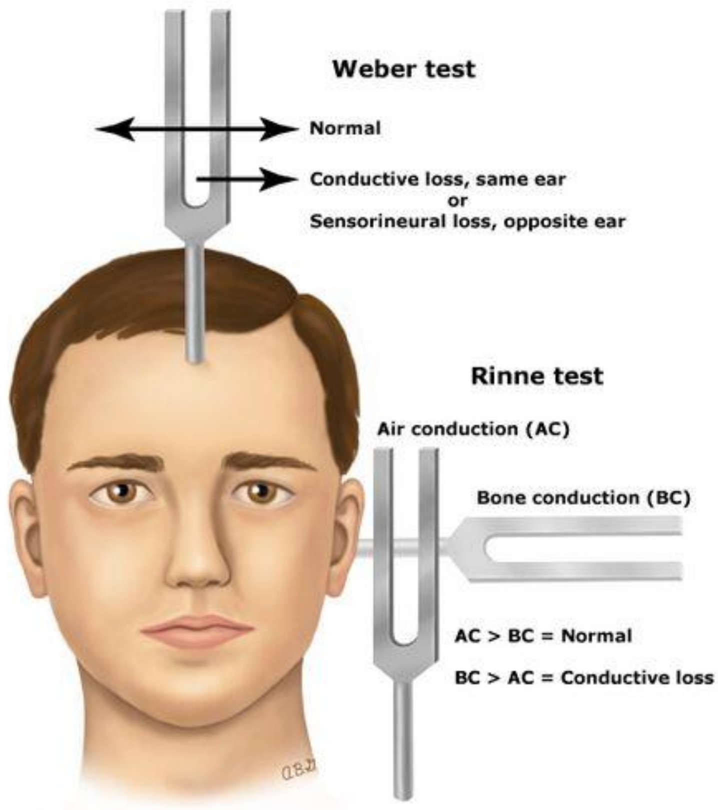
VESTIBULE EVOKED MYOGENIC POTENTIAL (VEMP)
Test for -1.Saccule 2.Inferior vestibular division of 8th CN
Indication - 1.Acoustic neuroma 2.Superior SCC dehiscence 3.Meniere's disease

AUDITORY PATHWAY - ECOLI MA
E- eight nerve
C- cochlear nucleus
O- olivary nucleus (superior)
L- lateral lemniscus
I- inferior colliculus
M- medial geniculate body
A- auditory pathway

HEARING TEST **NEETPG 24,FMGE 26,25**

Rinne test - comparison of Air Conduction and Bone Conduction

Normal = AC>BC	Rinne +ve
CHL= BC>AC	Rinne -ve
SNHL= AC>BC	Rinne +ve



Weber test

Normal	Sound heard → Centre of head
CHL	Sound heard → Poor ear
SNHL	Sound heard → Better ear

(Same side - CHL ,C/L side - SNHL)

Absolute bone conduction

Normal	bone conduction equal to examiner
CHL	bone conduction equal to examiner
SNHL	↓↓

Tunning fork Rinne test result **NEETPG 23**

256Hz	-ve	-ve	-ve
512Hz	+ve	-ve	-ve
1024Hz	+ve	+ve	-ve
CHL	15-30db	30-45db	45-60db

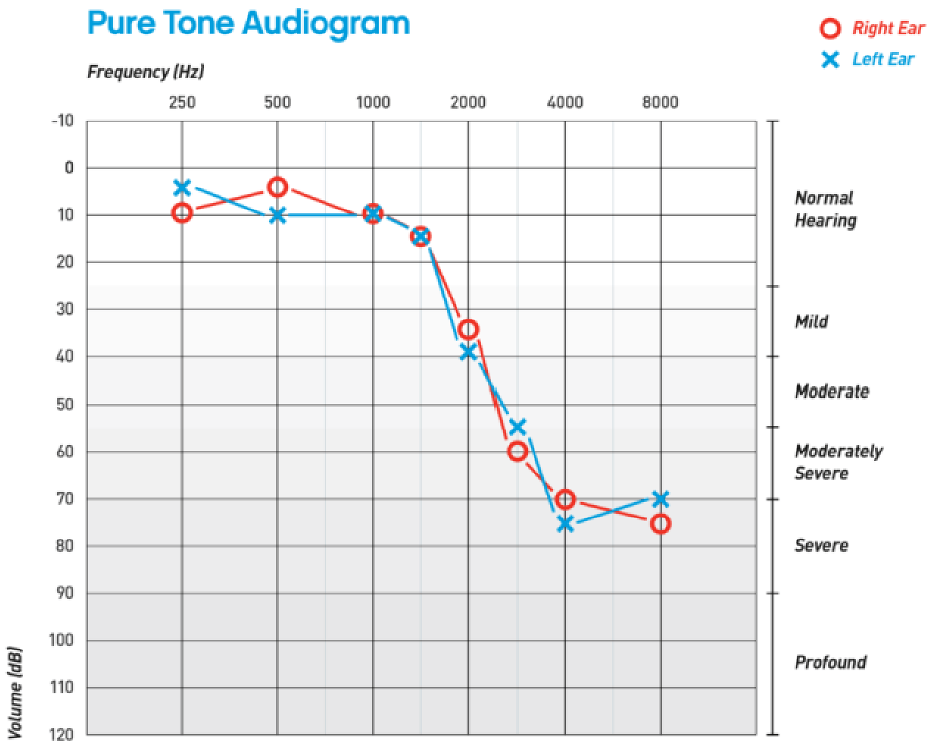
Frequently asked question:

Q)35-year-old patient presents with hearing loss. Clinical examination reveals a Rinne test where Air Conduction is greater than Bone Conduction in both ears. The Weber test lateralizes to the left ear. What is the most likely diagnosis?

a. Conductive hearing loss in left ear
b. Conductive hearing loss in the right ear
c. Sensorineural hearing loss in the left ear
d. sensorineural hearing loss in the right ear

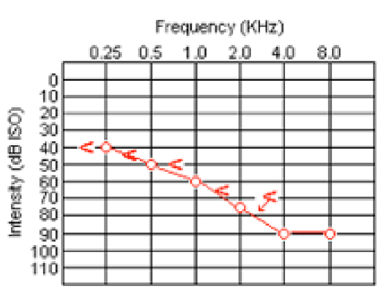
PURE TONE AUDIOMETRY **NEETPG 25**

Subjective test
Check both AC and BC level

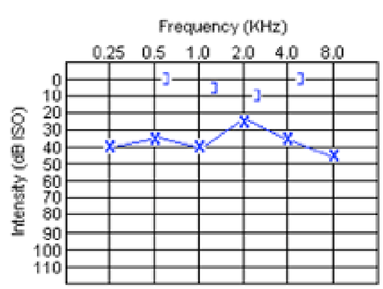


AB gap = CHL
Both AC and BC = SNHL
Both AC and BC poor with AB gap = mixed hearing loss

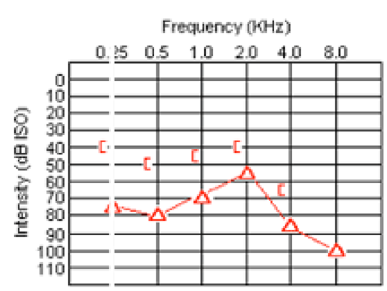
SENSORINEURAL HEARING LOSS



CONDUCTIVE HEARING LOSS

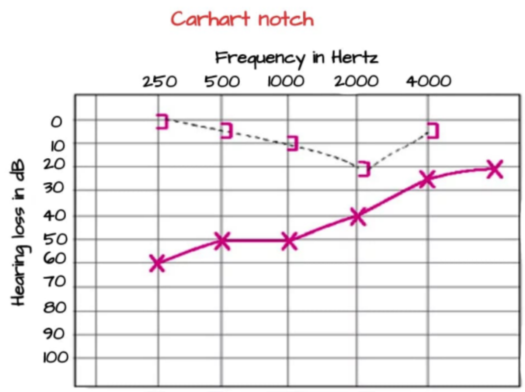


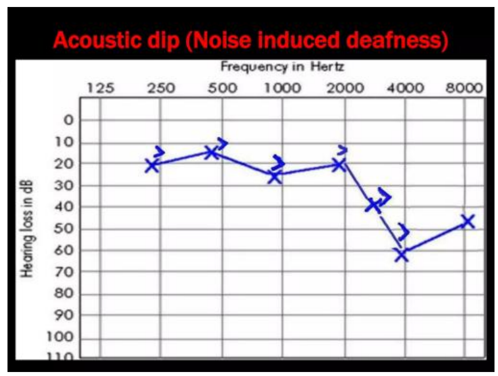
MIXED HEARING LOSS



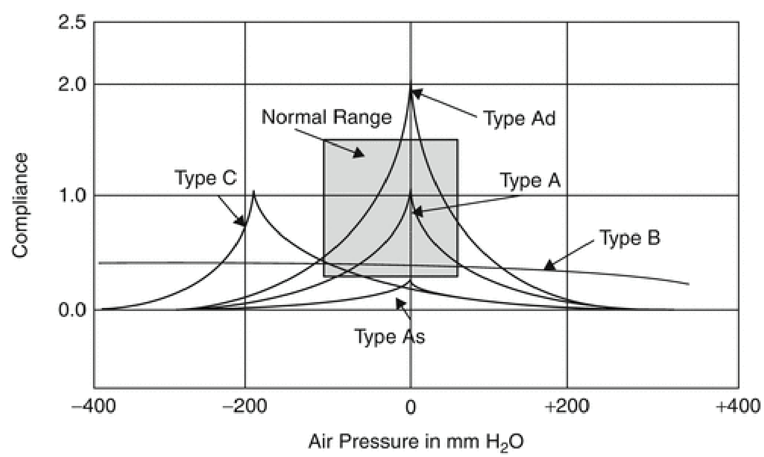
1.Otosclerosis
CHL - AB gap
Dip at 2000Hz in Bone conduction (carhart notch)

2.Noise induced hearing loss **FMGE 26**
SNHL
Dip at 4000 Hz in AC,BC → Acoustic Dip



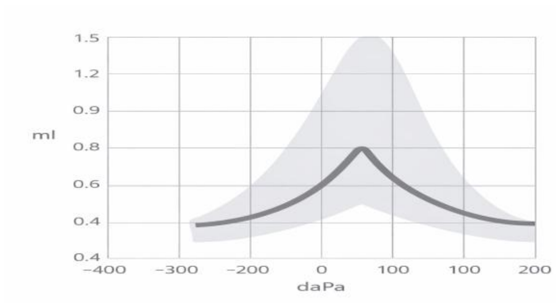


TYMPANOMETRY **FMGE 2025,26**
It shows 5 types of curves.
1. Type A is seen in normal people
2. Type B is seen in glue ear (flat curve)
3. Type C is seen ET dysfunction
4. Type As seen in otosclerosis
5. Type Ad is seen in ossicular dislocation



Frequently asked question:
Q) 5-year-old child presents with chronic nasal obstruction, mouth breathing, and hearing loss. Clinical examination suggests significant adenoid hypertrophy. Based on these findings, which type of tympanogram is most likely to be observed?
A. Type A Tympanogram
B. Type B Tympanogram
C. Type Ad Tympanogram
D. Type As Tympanogram

Q) Identify the type of tympanogram given in the image:



A. Type A Tympanogram.
B. Type B Tympanogram
C. Type Ad Tympanogram
D. Type As Tympanogram

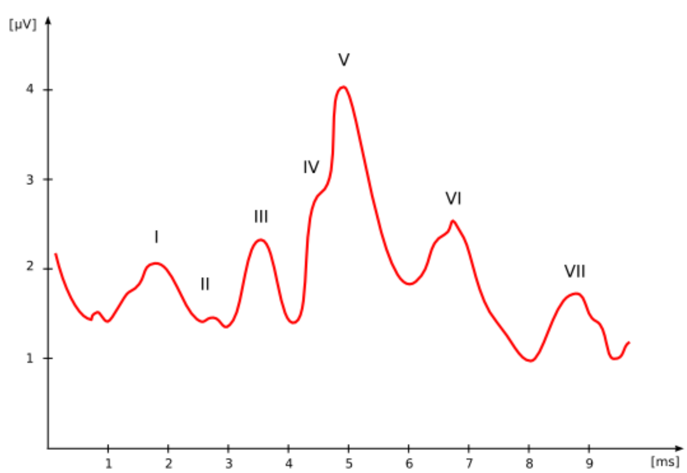
BRAINSTEM EVOKED RESPONSE AUDIOMETRY (BERA)

Objective test

Bera has 7 waves

Most important wave → wave V produced by lateral lemniscus

Gold standard → for hearing loss



Uses	Limitation
1.Pediatric hearing loss 2.Malingering 3.To differentiate between cochlear from retrocochlear disease	1.frequency non-specific 2.does not check auditory cortex

OTOACOUSTIC EMISSION (OAE) **FMGE 2024**

Sound to healthy cochlea → produces echoes
Echoes produced by outer hair cells

Best screening test for hearing Neonates → OAE High risk infants → BERA

OTOTOXICITY

Damage basal turn of cochlea (high frequency SNHL) in early stage

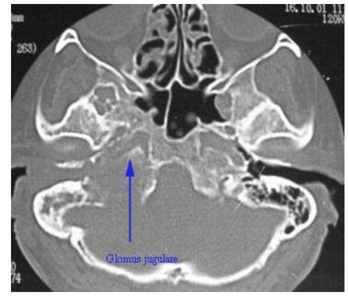
Drug induced hearing loss
1.Aminoglycoside → damage outer hair cells of cochlea
Amikacin streptomycin gentamycin
2.Furosemide
3.Chloroquine,Quinine

Dx - High frequency audiometry

GLOMUS JUGULARE **NEETPG 24,FMGE 2023**

Arise from glomus cells around jugular bulb
Benign,non capsulated,highly invasive tumor
c/f- pulsatile tinnitus
Red pulsatile mass in EAC which bleeds to touch

Brown sign → red mass in EAC blanches on siegelisation
Inv- CECT- phelp sign
Trt - surgery
Gamma knife



OTOSCLEROSIS **NEETPG 24,FMGE 2024,22**

Fixation of foot plate of stapes
C/F: young female with bilateral gradually progressive conductive hearing loss just marks OTOSCLEROSIS.
The patient hears better in a noisy area called: PARACUSIS WILLISII
HPE→ blue mantles seen in early stage
shows SCHWARTZ SIGN (flamingo pink appearance behind tympanic membrane) and TM is intact.
Site of origin - anterior to oval window

PTA shows dip at 2000 Hz in BC carhart's notch.
Tympanometry shows As curve.
NaF given in early stage of otosclerosis
Rx: TOC: Sx (stapedotomy/stapedectomy)
In this surgery we replace fixed stapes with artificial STAPES PISTON PROSTHESIS

ACOUSTIC NEUROMA(VESTIBULAR SCHWANNOMA)

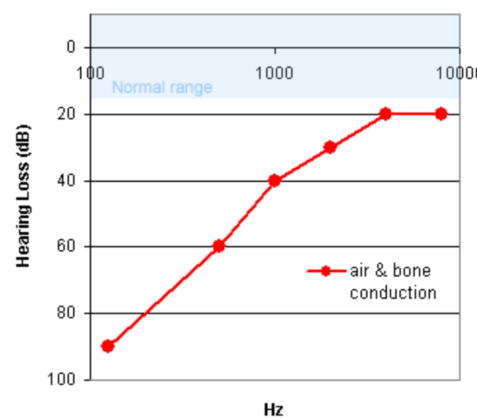
Benign tumor of 8th nerve It is m/c type of cerebellopontine angle brain tumor.
Most common site of origin → inferior vestibular
C/F: 1. U/L gradually progressive SNHL
2. Tinnitus
3. Imbalance
4. Patient has poor understanding of words & on increasing sound intensity understanding falls further this is called: ROLLOVER PHENOMENON

Best radiological investigation is gadolinium enhanced MRI.
Tone decay test → faster in retrocochlear hearing loss

Rx-
TOC- surgery
Gamma knife surgery

MENIERE’s DISEASE **FMGE 2025,24,22**
C/F: episodic disease
TRIAD:
1. Tinnitus
2. Vertigo
3. Hearing loss Patients fall down during an episode without turning unconscious. This is called: TUMARKIN'S CRISIS.

In between episodes:
1. The patient hears a loud sound as more loud this is called: RECRUITMENT PHENOMENON.
2. Patient gets vertigo on hearing loud sound this is called: TULLIO'S PHENOMENON.
3. The patient hears the same sound in 2 frequencies this is called: DIPLACUSIS.



Inv-
IOC: Electro-cochleography → SP/AP ratio > 30%
PTA - U/L low frequency SNHL → rising audiogram
Short increment sensitivity index → >70%

Trt-
Do PTA
1.Hearing preserved → do conservative management
Best option - vestibular neurectomy

2.No residual hearing → do Radial treatment
Labyrinthectomy
a.surgery
B.chemical (gentamicin)

BELL'S PALSY **FMGE 2025**
Idiopathic ,Sudden onset , CN VII involve
Lower neuron palsy, mostly U/L

c/f - Forehead muscles paralysed
Angle of mouth deviated to normal side
Loss of nasolabial fold

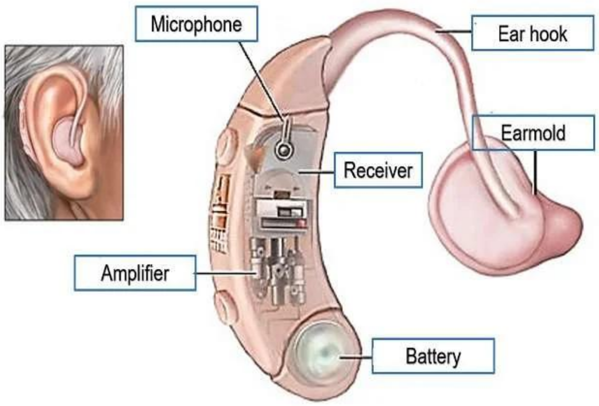
Hyperacusis - stapedial reflex gone

DOC- oral steroids

Acyclovir given to patient if present within 72 hours

HEARING DEVICES AND IMPLANTS

1.Hearing aid



2.Cochlear implant

Indication → B/L profound SNHL with no improvement seen after 6 months of hearing aid use

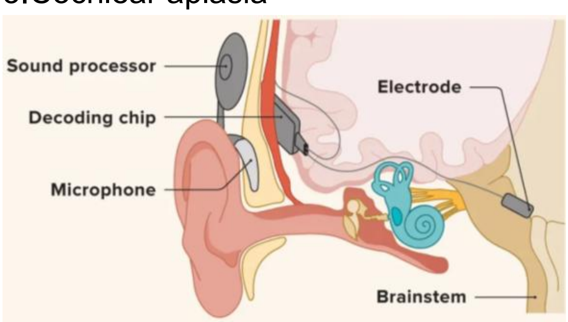
3.Auditory brainstem implant - ABI electrode placed in lateral recess of 4th ventricle

Indication -

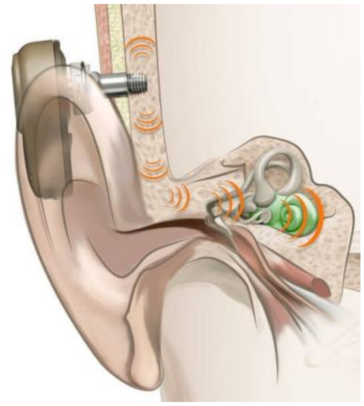
1.NF-2

2.Congenital aplasia of 8th CN

3.Cochlear aplasia



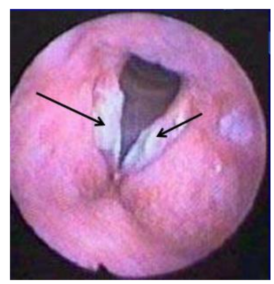
4.Bone anchoring hearing aid
Stimulate cochlea directly via bone conduction
Indication -
1.Anotia with hearing loss
2.EAC stenosis
3.U/L severe SNHL



LARYNX
CARTILAGES

Paired	Unpaired
Thyroid Cricoid Epiglottis	Arytenoid Cuneiform Corniculate

KERATOSIS LARYNX
Seen in smokers
It is premalignant disease
c/f- hoarse voice
Trt- surgery → stripping of mucosa



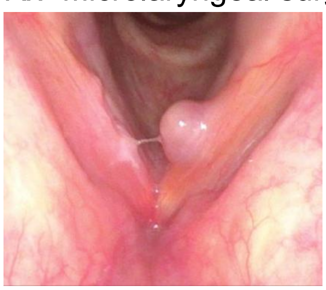
LARYNGOCELE
Abnormally dilated saccules →pierces thyrohyoid membrane
Seen in Trumpet,glass blowers
Bryce sign → sound of air leak when apply pressure on laryngocele
Dx- X-ray neck with valsalva manuvere
Rx- surgical excision

REINKE EDEMA **FMGE 2023,26**
B/L diffuse polyposis
Seen in smokers
c/f- hoarse voice
Trt - Sx → stripping of vocal cord

INTUBATION GRANULOMA
H/o of faulty intubation
Site- ant 2/3rd and post 1/3rd
Sx- MLS

VOCAL NODULE FMGE 2024,23
Cause- vocal abuse Bilateral
Site- junction of ant. 1/3rd and post 2/3rd of vocal cord.
C/F- hoarse voice
Rx- voice rest (speech therapy)

VOCAL POLYP
Unilateral Cause-Vocal abuse
Site- junction of ant 1/3rd and post. 2/3rd
Rx- Microlaryngeal surgery



JUVENILE PAPILLOMA OF LARYNX
Caused by HPV 6 & 11
Examination: Viral warts in the larynx- they can spread to trachea and bronchi
C/F: 4-6 year old child with chronic hoarseness of voice +/- Respiratory difficulty
Rx: Sx (MLS with co2 laser)
co2 laser is most commonly used laser in larynx surgery

PITCH DISORDER OF VOICE

PUBERPHONIA	**ANDROPHONIA**
High pitch voice in male Rx- speech therapy → Gutzmann manuvere ↓	Female with masculine voice Rx- Type IV thyroplasty (tightening of vocal cords)

If fails → do surgery → Type III thyroplasty (surgical loosening of vocal cord)	

LARYNGOMALACIA **FMGE 24,26**

Congenital laryngeal stridor → weakness of supraglottis
Most common disease of larynx
Most common cause of **neonatal inspiratory stridor**
On examination - omega shape epiglottis
Cry sound is normal
c/f- inspiratory stridor
Stridor ↑ on crying,omega sign improve in prone position

FMGE 2024

RHINOLALIA APERTA (HYPERNASALITY OF VOICE)	**RHINOLALIA CLAUSE (HYPONASALITY OF VOICE)**
Causes: 1. Cleft palate 2. Palatal palsy 3. Palatal perforation 4. Bifid uvula	Causes: 1. Nasal polyp 2. Sinusitis 3. Adenoid hypertrophy 4. Angiofibroma

TRACHEOSTOMY **NEETPG 24,FMGE 2022**

Mc site for tracheostomy is 2nd and 3rd tracheal rings called mid tracheostomy.
Higher tracheostomy- is done at 1st and 2nd tracheal ring done in CANCER LARYNX.
Tracheostomy can lead to -Mc - Hemorrhage
Air beneath the skin (surgical emphysema).
Apnea- its due to co2 washout.

MUSCLES OF LARYNX **NEETPG 24**

1.Abductor (1)	**2.Adductors (4)**	**3.Tensors (2)**
Posterior cricoarytenoid muscle	Thyroarytenoid Interarytenoid Lateral cricoarytenoid	Cricothyroid (main tensor) Vocalis muscle Function: Quality of voice

	Cricothyroid	

All these muscles lie inside larynx except CRICOTHYROID MUSCLE
All these muscles are supplied by RECURRENT LARYNGEAL NERVE (RLN) except CRICOTHYROID which is supplied by EXTERNAL BRANCH OF SUPERIOR LARYNGEAL NERVE

PHARYNX

ADENOID HYPERTROPHY **FMGE 2024,23**

Disease of school age children
Cause: recurrent infection - leads to more than physiological enlargement of adenoids.
C/F:School age child
B/L presentation,CHL due to glue ear
H/O mouth breathing
Adenoid face
1. Open mouth
2. Pinched nose
3. High palate
4. Malocclusion of teeth (teeth are not touching)

Rx: Surgery- Adenoidectomy
Method of Surgery is called: curettage Instrument: St. Clair Thomson Adenoid curette Position of patient during surgery: Rose position.

New methods of adenoidectomy
1.coblation
2.suction diathermy
3.microdebrier

ANGIOFIBROMA **NEETPG 25,FMGE 2025,24**

12-15 year old boy with Nasal mass + profuse epistaxis Mark- angiofibroma
It's a benign tumor Highly vascular
Site of origin- sphenopalatine foramen

Investigation-
1. CECT → Hollman Miller sign
2. ANGIOGRAPHY

Staging – FISCH/ Radkowski’s
Rx- Surgery
For tumors limited to nasopharynx and nose →
1.Endoscopic approach
2.wilson transpalatal approach

Extended to cheek
1.Lateral rhinotomy with medial maxillectomy approach

FMGE 25,26

RLN gone- B/L abductor palsy (RLN- recurrent laryngeal nerve)
Both vocal cords come in medial position.
Trt- type 2 Thyroplasty (Lateralisation of vocal cord)

RLN+ SLN gone- B/L adductor palsy
Both vocal cords come in lateral position.
Trt- type 1 Thyroplasty (medialisation of vocal cord) (sln-superior laryngeal nerve)

SLN gone- bowed down the vocal cord.
Difficulty in raising the pitch and tone of voice

FOREIGN BODY IN AIRWAY **NEETPG 22**
Most common in children
More common in Right bronchus
Bronchial foreign body → sudden bouts of cough,wheezing,↓air entry to lungs
Best inv- X-ray chest PA view with expiration phase
Laryngeal foreign body → do Heimlich maneuver → if fails do cricothyroidectomy

NASOPHARYNGEAL CARCINOMA (NPC) **NEETPG 25,23,FMGE 2025,24,22**
Hidden cancer (occult primary) Etiology: EB virus
Site of origin: fossa of ROSENMULLER (MC location for NPC)
Most common presentation: secondary neck node

NPC involves cranial nerves leads to TROTTER'S TRIAD:
1.Neuralgic pain in the temporoparietal area due to 5th nerve involvement.
2.Palatal palsy due to 10th nerve involvement.
3.Conductive hearing loss (unilateral)
IOC - CECT
Rx: Chemoradiation Radiotherapy.

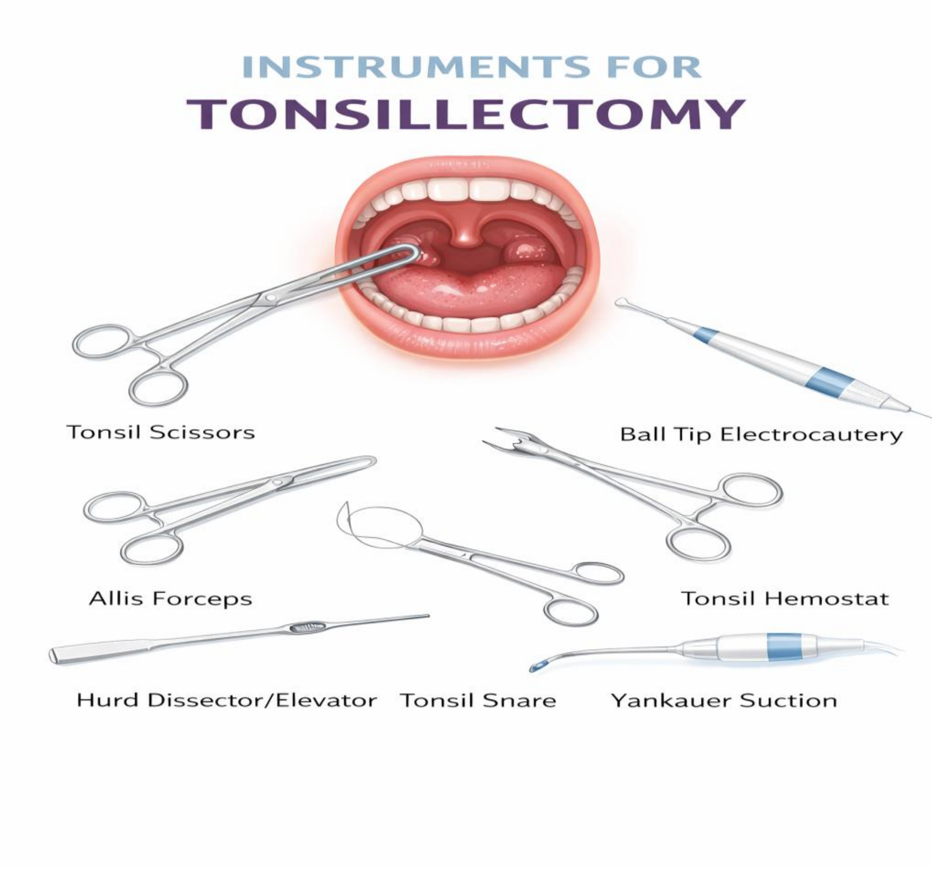
Most important source of blood supply of tonsil- Tonsillar branch of facial artery.
Venous drainage of tonsil: Paratonsillar vein - it is the main source of bleeding during Tonsillectomy

Bed of tonsil
Made by superior constrictor muscle
Structures in bed of tonsil

1) Styloid process
2) Glossopharyngeal Nerve

Elongated styloid process press the glossopharyngeal Nerve which leads to Throat pain referred to ear this phenomenon is called **Eagle syndrome or styalgia**

Q) Which artery lies one inch deep to the tonsil?
Answer- Internal carotid artery therefore pulsating tonsil can be a feature of ICA aneurysm
Tonsillectomy m/c method: Dissection and Snare method



HEMORRHAGE IN TONSILLECTOMY
Has 3 type:
1. Primary during Surgery
2. Reactionary - Post operation (It is within 24 hours of surgery.)
Cause: Slippage of ligature. It is an emergency
Rx: Immediate re-exploration.

3. Secondary- post op. After the 5th day of Surgery.
Cause: infection Mild bleeding only
Rx: IV antibiotics

QUINSY DISEASE/PERITONSILLAR ABSCESS	PARAPHARYNGEAL ABSCESS
Unilateral C/F: 1. Throat pain 2. Dysphagia 3. Trismus 4. Idiopathic edema of uvula 5. plummy voice Rx-I/D → remove tonsil after 6 weeks (interval tonsillectomy) 	Quinsy + outer neck swelling close to angle of mandible 

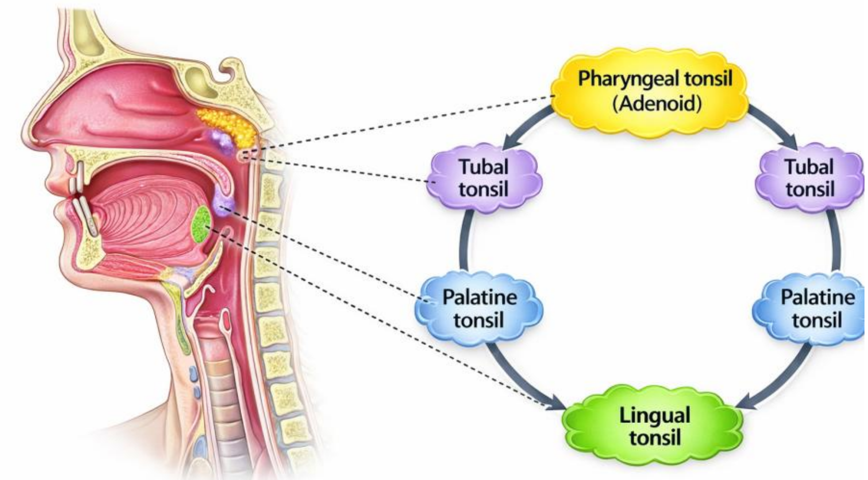
ACUTE RETROPHARYNGEAL ABSCESS

Pediatric emergency
U/L bulge on posterior pharyngeal wall
C/F:
1. Respiratory difficulty
2. Dysphagia
3. Drooling of saliva
X-ray neck will show: Widening of prevertebral shadow.
Rx-
1. Airway management
2. Per oral incision & drainage

LUDWIG'S ANGINA

It is infection of floor of mouth (submandibular space)
Cause:
1. Dental infection
C/F:
After dental infection patient develops:
1. Chin swelling
2. Trismus
Bacteria: streptococci + anaerobes (mixed)
Rx: External incision & Drainage.

Waldeyer Ring



JUVENILE PAPILLOMA OF LARYNX
Caused by HPV 6 & 11.
Examination:
Viral warts in the larynx - they can spread to trachea and bronchi.
C/F: 4-6 year old child with chronic hoarseness of voice +/- Respiratory difficulty.
Rx: Sx (MLS with co2 laser) co2 laser is most commonly (used laser in larynx surgery)

CHOANAL ATRESIA **FMGE 2024**
Persistence of buccopharyngeal membrane
Airway emergency ? Neonates are obligatory,nasal breathers
c/f- Blue baby turns pink on crying
Immediate trt - put wide bore nipple in child mouth
McGroven Technique → if fails do tracheostomy → do recanalisation surgery after 1 yr of age

CANCER LARYNX STAGING **NEETPG 25,23FMGE 26,23**

T1 – Only 1 structure involved
T2 – More than 1 named structure involved
T3 – Vocal cord is fixed (immobile)
3 Ps- invasion of preglottic,paraglottic space,posterior cricoid invasion
T4 – Invasion of thyroid cartilage

N1 - Node less than 3cm
N2 - Node 3-6cm in size
N3- Node> 6cm in size

Rx:
T1 & T2: Radiotherapy
T3 & T4: Total laryngectomy (by gluck sorenson incision) followed by Radiotherapy
Second best option concurrent chemoradiation for T3 &T4

Now days TOC of T1 Glottic cancer – Laser Sx > radiotherapy
After a total laryngectomy patient has Permanent tracheostomy.

Stage 1	**T1N0 M0**
Stage 2	**T2 N0 M0**
Stage 3	**T3 N1 M0**
Stage 4	**T4 N2 M1**

Frequently asked question:
Q)CT scan of neck of a patient of carcinoma larynx has been found to have the involvement of epiglottis, right ventricular band(false vocal cord)and invading into paraglottic space. What is the ideal treatment of this patient?
a. Radiotherapy
<u>b. Total laryngectomy + radiotherapy</u>
c. Chemotherapy
d. Chemoradiation
Explanation: Since the stage is T3 as more than 2 structures involved so Treatment should be Chemoradiation as per new guidelines.

Q)A laryngeal carcinoma patient has involvement of

epiglottis and right aryepiglottic folds. Both Vocal cords are normal and mobile.
What is the ideal treatment of this patient?
<u>a. Horizontal Partial Laryngectomy</u>
b. Total laryngectomy + radiotherapy
c. Vertical Partial Laryngectomy
d. Chemoradiation
Explanation: Since the stage is T2 supraglottic cancer so ideal treatment should be Radiotherapy but in options there is no such option so we can go for partial Laryngectomy.

Q)58-year-old man, chronic smoker, presents with progressive hoarseness of voice and throat discomfort for 4 months. Laryngoscopy reveals a laryngeal growth, and neck examination shows a single ipsilateral, mobile cervical lymph node. Biopsy confirms squamous cell carcinoma of the larynx. What is the most appropriate treatment?
a. Radiotherapy alone
<u>b. Chemoradiotherapy</u>
c. Partial laryngectomy
d. Total laryngectomy
Explanation: Since there is involvement of lymph nodes so it is N2 stage 3 so Treatment should be Chemoradiation

FMGE 2025

ACUTE EPIGLOTTITIS	**ACUTE LARYNGOTRACHEOBRONCHITIS(CROUP)**
Infection of supraglottis	Infection of the complete airway but subglottis is the most affected part
Cause- Streptococcus Pneumoniae	Causes- Parainfluenza virus
C/f- 1. Inspiratory stridor 2. Respiratory distress 3. High fever 4. Drooling of saliva 5. Hot potato (plummy voice) 6. Child sits bending forward (tripod position	C/f- 1. Biphasic stridor 2. Respiratory distress 3. Low fever (viral infection gives low fever) 4. Barking cough
X-ray neck will show THUMB sign (swollen epiglottis)	X-ray neck shows- STEEPLE sign. It is narrowing of subglottis

	
Rx- 1. 1st treatment is a secure airway by intubation 2. Antibiotics	Rx- 1. Airway management 2. Bronchodilator 3. Steroids

NOSE

DEVELOPMENT OF SINUSES **NEETPG 24,FMGE 2025**

Radiologically sinuses appear in this M-E-S-F

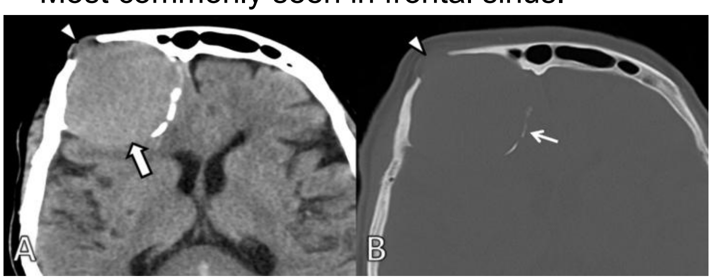
Maxillary sinus Ethmoid sinus Sphenoid sinus Frontal sinus

S-appear at 4 year of age

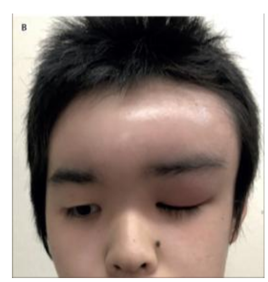
F-appear at 6 year of age

COMPLICATION OF SINUSITIS **NEETPG 23**

1. Orbital infection
   Orbital infection most seen in ethmoid sinusitis.
2. Mucocele formation
   Big sinus
   Most commonly seen in frontal sinus.



3. Pott's puffy tumor Frontal sinusitis - lead to - Frontal osteomyelitis - leads to subperiosteal frontal abscess formation (pott's puffy tumor)

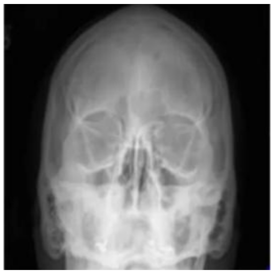
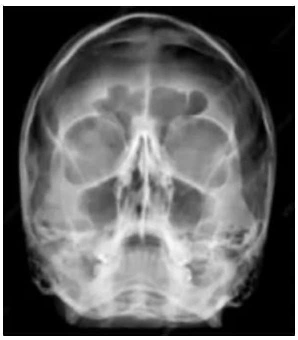


Note:

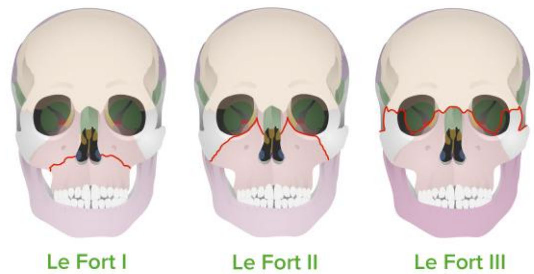
Artery of epistaxis- Sphenopalatine artery

1st Artery to ligate in epistaxis is- sphenopalatine artery If failed then ligate maxillary

artery ligation- if its failed- external carotid artery ligation- If its failed then ethmoidal artery ligation
Laryngeal crepitus- it is the clicking sensation felt when the larynx is moved over cervical vertebrae- present in normal people. Absent in posterior Cricoid carcinoma.

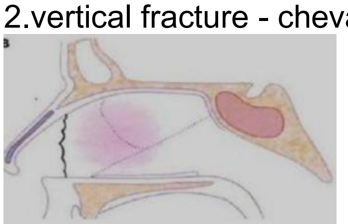
CALDWELL VIEW **FMGE 2022**
Occipitofrontal view → frontal sinus assessment

WATERS'S VIEW
Occipitomeatal view → maxillary sinus assessment

PIERRE'S VIEW
Waters view take open mouth → sphenoid sinus assessment

MIDFACE FRACTURE (MAXILLA)



NEETPG 25

<table>
<tr><th>NASAL SEPTUM FRACTURE</th><th>NASAL BONE FRACTURE</th></tr>
<tr><td>2 types
1.Horizontal fracture - jarjaway #

2.vertical fracture - chevallet #


Trt - close reduction of #
By ASCH FORCEPS

Asch Nasal Septum Forceps - Curved</td><td>Most common bone to be # on face

Rx- closed reduction
Walsham forceps

Walsham Nasal Septum Forceps (R-L)

1.No swelling → immediate # reduction
2.swelling + → wait for 7 days to subside swelling → then do reduction</td></tr>
</table>

CSF RHINORRHEA **FMGE 2025,24**

Cause:

1. Nasal surgery

2. Trauma -leads to skull base fracture- leads to traumatic CSF leak-blood mixed CSF-on filter paper-dry it -we can see "target sign" (halo sign) (double halo sign)
These signs are not seen in every CSF Rhinorrhea,seen only in trauma.

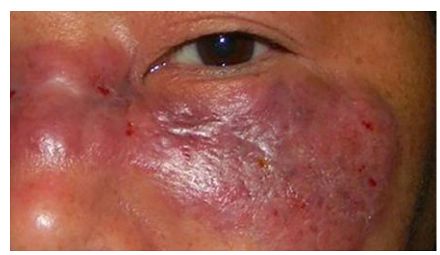
The most common site of CSF rhinorrhea is CRIBRIFORM PLATE.
Initial management – Bed rest with head elevation.
To prevent Secondary infection – antibiotic therapy with acetazolamide
NOTE – Vitamin D deficiency leads to Laryngismus stridulus & is associated with Rickets.

STRUCTURE DRAINING INTO NASAL CAVITY **FMGE 2024,23**

1.Nasolacrimal ducts drain into inferior meatus.
2.Maxillary sinus, frontal sinus & anterior ethmoid air cell drain into- Middle meatus.
3.Posterior ethmoid air cells drain into -Superior meatus.
4.Sphenoid sinus drain into - Sphenoethmoidal recess.

MUCORMYCOSIS **NEETPG 22,FMGE 2025,23,22**

1. It is a fungal infection of the nose by a mucor group of fungi.
2. It is seen in young, diabatic, HIV positive patients.



Clinical feature:
Mucor is angioinvasive fungus it enter to orbit and brain
1. Blackish mass in nose
2. Blackish discoloration around eye BLACK
Dx- biopsy with histological examination
Rx: DOC: amphotericin-B

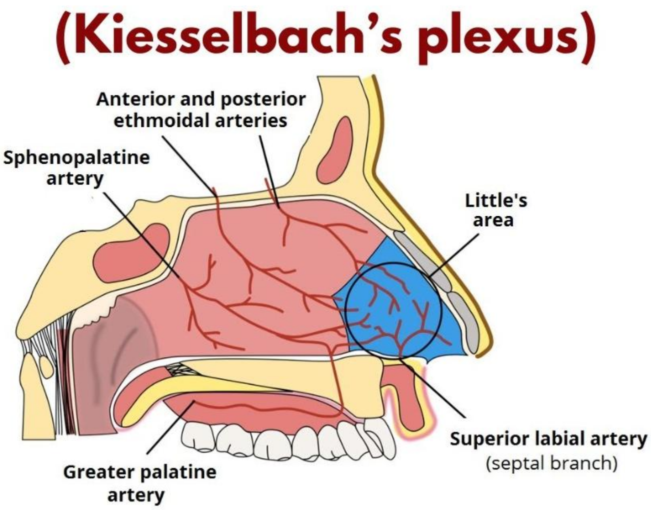
EPISTAXIS

The most common artery to bleed is the sphenopalatine artery.

Most common areas lie at anteroinferior part of the nasal septum.

It has Kiesselbach's plexus of 4 arteries (we have a total of 5 arteries SP,GP,SL,AE, PE,).

PE (posterior ethmoidal artery) is not a part of Kiesselbach's plexus.



Causes:

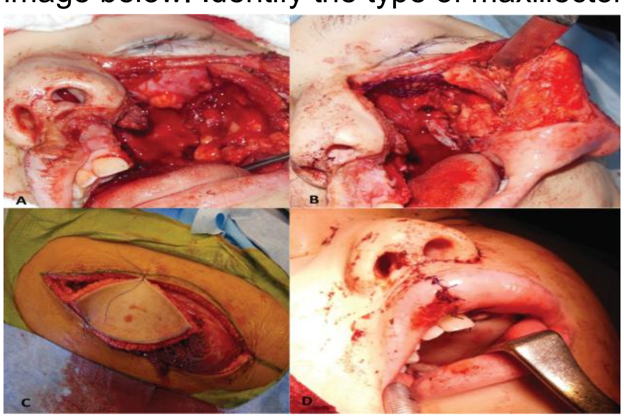
1. M/c/c finger nail trauma (nose picking)
2. Accidental trauma
3. Hypertension (causes posterior epistaxis)

Rx:

1. Pinch the nose for 3-5 minute
2. Chemical cauterization of little area with AgN03.
   AgNo3 is a cauterization agent that will burn that area and bleeding will stop.
3. Anterior nasal packing (on both side) (if this failed)
4. Posterior + anterior nasal packing.

Frequently asked question:

Q) A patient with maxillary cancer underwent the surgical procedure as shown in the image below. Identify the type of maxillectomy performed.



<u>A.Total Maxillectomy</u>

B. Medial Maxillectomy

C. Subtotal maxillectomy

D. Infrastructure maxillectomy

TUMORS OF PARANASAL SINUSES AND NOSE

Involve-Maxillary> ethmoid

Risk factor:

1. Nickel squamous cell carcinoma
2. Wood dust adenocarcinoma of ethmoid

If the cancer is on the maxillary sinus then it can easily spread to the eye because the maxillary sinus is just below the eye. To check cancer goes to eye or not we use line called: ohngren's line

Ohngren's line: medial canthus to angle of mandible.

If the cancer above this line then it will easily go to the eye. It will have a poor prognosis.

Tx: Surgery (total maxillectomy) by weber ferguson approach F/b radiotherapy

THERE ARE 2 TYPES OF POLYPS

1.Antrochoanal polyp **FMGE 2023**

Arises from maxillary sinus and goes posteriorly towards choana.

Better seen on posterior rhinoscopy.

It is more common in children

It is Single polyp and u/l polyp.

Tx: Sx endoscopic polypectomy or FEES.

2.Ethmoidal polyp (nasal polyp) **FMGE 2023**

Most common (90%)

Arise from ethmoid air cell

It is more common in adult

It is due to chronic allergy

It is multiple and bilateral

Tx: Topical corticosteroid nasal spray. Eg. Fluticasone

SAMPTER's TRIAD **NEETPG 22**

1. Nasal polyp
2. Asthma
3. Allergy to NSAIDs (aspirin)

INVERTED PAPILLOMA OF NOSE (RINGERTZ TUMOR) **FMGE 2023**

More common in Male, grow inwards,benign and invasive

Arise from Lateral wall of nose

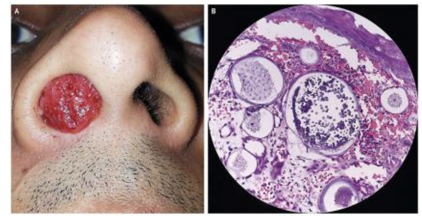
NASAL POLYP

It is a prolapsed pedunculated oedematous mucosa of sinuses.

Etiology:

1. 2. Chronic infection / allergy for the last few years.

It can cause chronic inflammation that leads to oedema - polyp.



RHINOSPORIDIOSIS
Caused by- rhinosporidium Seeberi → Aquatic protozoa
Risk factor- bathing in ponds
Examination shows- mulberry or strawberry like nasal mass + epistaxis
Rx- Sx excision of mass with electrocautery of base(burn the base) followed by dapsone(doc)

RHINOSCLEROMA (woody nose) **FMGE 26**
Cause by- K. Rhinoscleromatis
It has 3 stages:

1.Atrophic stage just like atrophic rhinitis: Roomy nasal cavity and crust
2.Granulomatous stage that leads to a hard external nose (woody nose).
3.Stage of fibrosis
Dx- biopsy It shows Russel bodies & Mikulicz cells.
Rx- Tetracycline and Streptomycin

NOTE - **FMGE 2024**
1.Ant. Ethmoidal sinus drain into – Middle Meatus
2.Ant. Ethmoidal Neuralgia AKA Sluder's Neuralgia – Compression of Anterior Ethmoidal Nerve d/t middle(Most often) /inferior turbinate

ATROPHIC RHINITIS **FMGE 2025,24**
Caused by klebsiella ozaenae

Progressive atrophy of mucosa,submucosa,underlying bones of nasal cavity
o/e- 1.roomy nasal cavity 2.Shrunken turbinates 3.Merciful anosmia
Trt- alkaline nasal douching
1.NaHco3 2.Sodium bicarbonate 3.NaCl
Surgery → modified young operation